

Median

Example:

- Obtain the median for the following frequency distribution:

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6

Solution:

For a discrete frequency distribution, the median is the value of x corresponding to the $(\frac{N+1}{2})^{\text{th}}$ observation, where $N = \sum f$. We first calculate the **Cumulative Frequency** ($c. f.$).

x	f	$c. f.$
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120
Total	N = 120	

Total frequency: $N = 120$.

$$\text{Median Position} = \frac{N+1}{2} = \frac{120+1}{2} = 60.5.$$

The $c. f.$ just greater than 60.5 is 65, which corresponds to $x = 5$.

$$\text{Median} = 5$$

Example:

- Find the median wages of the following distribution:

Wages	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000
No. of workers (f)	3	5	20	10	5

Solution:

For a grouped frequency distribution, the median is calculated using the formula:

$$M = L + \left(\frac{\frac{N}{2} - c.f.p}{f_M} \right) \times h$$

First, we find N and $N/2$, and then calculate $c.f.$ to find the median class.

Wages	f	$c.f.$
2000-3000	3	3
3000-4000	5	8
4000-5000	20	28
5000-6000	10	38
6000-7000	5	43
Total	N = 43	

$N = 43$. $N/2 = 43/2 = 21.5$.

The $c.f.$ just greater than 21.5 is 28, so the **Median Class** is **4000 – 5000**.

- $L = 4000$ (Lower limit of the median class)
- $c.f.p = 8$ (c.f. of the preceding class)
- $f_M = 20$ (Frequency of the median class)
- $h = 1000$ (Class size)

$$M = 4000 + \left(\frac{21.5 - 8}{20} \right) \times 1000$$

$$M = 4000 + \left(\frac{13.5}{20} \right) \times 1000$$

$$M = 4000 + 0.675 \times 1000$$

$$M = 4000 + 675 = \mathbf{4675}$$

The median wage is **Rs. 4675**.

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Mode

Example:

- Calculate the mode from the following frequency distribution:

Size (x)	4	5	6	7	8	9	10	11	12	13
Frequency (f)	2	5	8	9	12	14	14	15	11	13

Solution:

For a discrete distribution, the mode is the value of x that has the highest frequency. The highest frequency is **15**, which corresponds to the size **$x = 11$** .

$$\text{Mode} = \mathbf{11}$$

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Example:

- Find the mode of the following:

Marks	0-5	6-10	11-15	16-20	21-25	26-30	31-35	36-40	41-45
No. of candidates (f)	7	10	16	32	24	18	10	5	1

Solution:

The classes are **discontinuous** (inclusive). We first make them continuous (exclusive) by subtracting 0.5 from the lower limit and adding 0.5 to the upper limit. (e.g., 0 – 5 becomes –0.5 – 5.5).

The formula for Mode in grouped data is:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

1. **Identify Modal Class:** The highest frequency is $f_1 = 32$, so the modal class is 16 – 20

2. **Continuous Class:** The continuous modal class is 15.5 – 20.5.

3. **Parameters:**

- $L = 15.5$ (Lower limit of the modal class)
- $f_1 = 32$ (Frequency of the modal class)
- $f_0 = 16$ (Frequency of the preceding class, 11-15)
- $f_2 = 24$ (Frequency of the succeeding class, 21-25)
- $h = 5$ (Class size: 20.5 – 15.5)

$$\text{Mode} = 15.5 + \left(\frac{32 - 16}{2(32) - 16 - 24} \right) \times 5$$

$$\text{Mode} = 15.5 + \left(\frac{16}{64 - 40} \right) \times 5$$

$$\text{Mode} = 15.5 + \frac{16}{24} \times 5$$

$$\text{Mode} = 15.5 + \frac{2}{3} \times 5 \approx 15.5 + 3.3333 = \mathbf{18.83}$$

Example:

- Define the Mode and calculate Mode for the distribution of monthly rent Paid by Libraries in Karnataka:

Monthly rent	500-1000	1000-1500	1500-2000	2000-2500	2500-3000	3000 and above
No. of Library (f)	5	10	8	16	14	12

Solution:

The **Mode** is defined as the value that occurs most frequently in a distribution. For grouped data, it is the value with the greatest density (highest frequency).

We use the same formula as above:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

1. **Identify Modal Class:** The highest frequency is $f_1 = 16$, so the **Modal Class** is **2000 – 2500**.

2. **Parameters:**

- $L = 2000$ (Lower limit of the modal class)
- $f_1 = 16$
- $f_0 = 8$ (Frequency of the preceding class, 1500-2000)
- $f_2 = 14$ (Frequency of the succeeding class, 2500-3000)
- $h = 500$ (Class size)

$$\text{Mode} = 2000 + \left(\frac{16 - 8}{2(16) - 8 - 14} \right) \times 500$$

$$\text{Mode} = 2000 + \left(\frac{8}{32 - 22} \right) \times 500$$

$$\text{Mode} = 2000 + \frac{8}{10} \times 500$$

$$\text{Mode} = 2000 + 0.8 \times 500 = 2000 + 400 = \mathbf{2400}$$

The modal monthly rent is **Rs. 2400**.

Mean, Median, and Mode

Example:

- Calculate the mean, median and mode of the following data:

Wages (in Rs)	0-20	20-40	40-60	60-80	80-100	100-120	120-140
No. of Workers (f)	6	8	10	12	6	5	3

Solution:

We calculate the midpoint (m) and cumulative frequency ($c. f.$) to find the measures of central tendency.

Wages	f	m	$f \cdot m$	$c. f.$
0-20	6	10	60	6
20-40	8	30	240	14
40-60	10	50	500	24
60-80	12	70	840	36
80-100	6	90	540	42
100-120	5	110	550	47
120-140	3	130	390	50
Total	$N = 50$		$\sum(f \cdot m) = 3120$	

1. Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum(f \cdot m)}{N} = \frac{3120}{50} = 62.4$$

2. Median (M)

$N/2 = 50/2 = 25$. The *c. f.* just greater than 25 is 36, so the **Median Class** is **60 – 80**.

- $L = 60, h = 20, f_M = 12, c. f.p = 24$.

$$M = L + \left(\frac{\frac{N}{2} - c. f.p}{f_M} \right) \times h = 60 + \left(\frac{25 - 24}{12} \right) \times 20$$

$$M = 60 + \frac{1}{12} \times 20 \approx 60 + 1.6667 = \mathbf{61.67}$$

3. Mode (Z)

The highest frequency is 12, so the **Modal Class** is **60 – 80**.

- $L = 60, h = 20, f_1 = 12, f_0 = 10, f_2 = 6$.

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h = 60 + \left(\frac{12 - 10}{2(12) - 10 - 6} \right) \times 20$$

$$\text{Mode} = 60 + \frac{2}{24 - 16} \times 20 = 60 + \frac{2}{8} \times 20 = 60 + 5 = \mathbf{65.0}$$

Final Results:

- Mean $\approx$ **Rs. 62.40**
- Median $\approx$ **Rs. 61.67**
- Mode = **Rs. 65.00**